Mathematical Abstract of a DCT-QIM Watermarking System with RS Coding and Soft Beam Decoding

Implementation: trajectory_watermark_masterkey_rs_forney_softbeam

Abstract

This note gives a concise mathematical description of a watermarking pipeline that embeds a short payload into the mid-frequency DCT coefficients of image blocks using Quantization Index Modulation (QIM), protects the payload with Reed–Solomon (RS) channel coding over GF(2^8), and recovers candidates via soft symbol scoring and a byte-level beam search. A case-insensitive normalized Levenshtein similarity is used to report a *match percentage* between a user-supplied ideal string and the recovered candidate; this percentage provides a robust, interpretable indicator of correspondence even when bit-level integrity checks fail or the file has been heavily modified.

1 Signal model

Let $I : \Omega \subset \mathbb{Z}^2 \to [0, 255]$ denote the luminance component Y of an image. Partition I into non-overlapping 8×8 blocks $B_{m,n}$. For each block compute the 2-D Discrete Cosine Transform (DCT):

$$C_{u,v}^{(m,n)} \;=\; \frac{1}{4}\,\alpha(u)\alpha(v)\sum_{x=0}^{7}\sum_{y=0}^{7} I_{m,n}(x,y)\cos\!\left(\frac{(2x+1)u\pi}{16}\right)\cos\!\left(\frac{(2y+1)v\pi}{16}\right),$$

where

$$\alpha(k) = \begin{cases} 1/\sqrt{2}, & k = 0,\ 4pt]1, \\ k \geq 1. \end{cases}$$

Fix a mid-frequency coordinate $(u^\star, v^\star)$ (implementation: zigzag index 12, corresponding to $[2, 2]$). Denote the coefficient of interest by $c = C_{u^\star,v^\star}^{(m,n)}$.

2 Payload coding and symbol mapping

Let the user payload be a short byte string M. Apply Reed–Solomon encoding over GF(2^8) to produce a codeword

$$E = (e_0, e_1, \ldots, e_{N-1}) \in \{0, \ldots, 255\}^N,$$

with $N = 32$ in the described implementation (16 data bytes + 16 parity bytes). Each byte e_i is expanded into four base-4 symbols using MSB-first ordering:

$$e_i \mapsto (s_{4i}, s_{4i+1}, s_{4i+2}, s_{4i+3}), \qquad s_j \in \{0, 1, 2, 3\}.$$

Concatenate these symbols to obtain a symbol sequence $\{s_k\}_{k=0}^{K-1}$.

3 QIM embedding

Quantization Index Modulation (QIM) embeds a symbol $s \in \{0, \ldots, V-1\}$ into coefficient c by shifting c to the nearest center of a uniform partition with step Δ and V sublevels:

$$\mathrm{embed}(c; s) \;=\; \left\lfloor \frac{c}{\Delta} \right\rfloor \Delta \;+\; \left(s + \tfrac{1}{2}\right) \frac{\Delta}{V}.$$

In the implementation $V = 4$ and Δ is a positive constant (e.g. $\Delta = 120$). After modifying $C_{u^\star, v^\star}$ the inverse DCT (IDCT) reconstructs spatial samples; these are clamped to the valid range $[0, 255]$.

To increase robustness, each symbol s_k is embedded redundantly across R spatially distributed blocks (replication factor R), and a small number of pilot slots with strong energy are inserted for synchronization and coarse detection.

4 Extraction: soft scoring and beam search

During extraction, for each slot k the algorithm inspects the R replicated blocks and computes an estimated symbol for each block by selecting the QIM level whose center is closest to the observed coefficient. Aggregating the R estimates yields counts or soft frequencies; normalize these to obtain soft probabilities

$$p_k(s) \;=\; \Pr(\text{symbol} = s \mid \text{observations for slot } k), \qquad s \in \{0, 1, 2, 3\},$$

with $\sum_s p_k(s) = 1$.

Group symbols in quadruples (MSB-first) to form candidate bytes. For a candidate byte formed from symbols $(s_k, s_{k+1}, s_{k+2}, s_{k+3})$ define a multiplicative score

$$\mathrm{score}(\text{byte}) \;\propto\; \prod_{j=0}^{3} p_{k+j}(s_{k+j}).$$

A beam search of width W keeps the top W partial byte sequences by score; final candidate byte sequences of length N are submitted to Reed–Solomon decoding (Berlekamp–Massey for error locator, Chien search for roots, Forney formula for error magnitudes). If RS decoding succeeds, a CRC8 check on the payload is used to validate integrity.

If RS decoding fails for beam candidates, a fallback uses majority voting on slot symbols and MSB-first unpacking to produce a best-effort candidate.

5 Levenshtein similarity and match percentage

Let s_{ideal} be the user-provided ideal string and s_{rec} the recovered string candidate. Compute the Levenshtein distance $d_L(s_{\text{ideal}}, s_{\text{rec}})$, the minimum number of insertions, deletions, and substitutions required to transform s_{ideal} into s_{rec}. Normalize the distance to obtain a similarity score:

$$\mathrm{sim}(s_{\text{ideal}}, s_{\text{rec}}) \;=\; 1 - \frac{d_L(s_{\text{ideal}}, s_{\text{rec}})}{\max\{|s_{\text{ideal}}|,\, |s_{\text{rec}}|\}},$$

with the convention that sim $= 1$ when both strings are empty. The reported match percentage is

$$\mathrm{match\%} \;=\; 100 \cdot \mathrm{sim}(s_{\text{ideal}}, s_{\text{rec}}),$$

formatted with one decimal digit. The comparison is *case-insensitive*: both strings are converted to lower case before computing d_L.

6 Interpretation and robustness

- **Spatial redundancy:** Replicating each symbol across R blocks and distributing slots across the image reduces the probability that local distortions (noise, cropping, localized filtering) simultaneously corrupt all replicas of a symbol.
- **Channel coding:** Reed–Solomon coding over $GF(2^8)$ corrects burst and random byte errors; combined with beam search, it enables recovery of plausible candidates even when many symbol estimates are noisy.
- **Match percentage significance:** The match percentage is a robust, interpretable indicator of correspondence between the ideal and recovered strings. Even when CRC verification fails or RS decoding is imperfect, a high match percentage (for example $70\% - 95\%$) signals a strong semantic match and suggests the watermark persists despite heavy modifications (resampling, aggressive compression, partial cropping). Low values indicate weak or absent correspondence.
- **Practical thresholds:** Thresholds for declaring a positive detection depend on application risk tolerance; near 100% indicates near-perfect recovery, intermediate values indicate partial recovery useful for forensic or probabilistic detection.

Summary

The described pipeline integrates DCT-domain QIM embedding, RS channel coding, and soft decoding with byte-level beam search to maximize recoverability of short payloads under common image and video distortions. The normalized Levenshtein-based match percentage provides a case-insensitive, quantitative metric that meaningfully quantifies correspondence even when bit-level integrity checks fail, enabling reliable detection and assessment of watermark presence in heavily modified files.